Dependency Extraction of OpenCV

November 26, 2025

SourceArchitects Group Members:

- Sheida Khodabakhshian Khansari
- Vrinda Ashwar
- Matthew Lee
- Werner Bagu-Paiz
- William Moran
- Stefewn Johnson

Dependency Extraction Techniques

- Understand[1]
- Include
- srcML[2]

Quantitative Analysis

TA Files with Extracted Dependencies

cleaned_understand_dependencie

File Edit View

```
FACT TUPLE :
$INSTANCE modules/gapi/include/opencv2/gapi/render/render.hpp cFile
$INSTANCE modules/core/test/test_concatenation.cpp cFile
$INSTANCE modules/ml/test/test_main.cpp cFile
$INSTANCE modules/features2d/src/kaze/nldiffusion_functions.cpp cFile
$INSTANCE modules/core/include/opencv2/core/check.hpp cFile
$INSTANCE modules/photo/test/test_cloning.cpp cFile
$INSTANCE modules/video/include/opencv2/video/detail/tracking.detail.hpp cFile
$INSTANCE build/3rdparty/ippicv/ippicv_lnx/iw/include/iw/iw_image_transform.h cFile
$INSTANCE build/modules/core/has_non_zero.simd_declarations.hpp cFile
$INSTANCE modules/highgui/test/test_main.cpp cFile
$INSTANCE modules/core/include/opencv2/core/opengl.hpp cFile
$INSTANCE modules/gapi/src/executor/gexecutor.hpp cFile
$INSTANCE modules/objdetect/src/barcode_decoder/ean8_decoder.hpp cFile
$INSTANCE modules/features2d/test/test_matchers_algorithmic.cpp cFile
$INSTANCE modules/features2d/test/test_descriptors_invariance.impl.hpp cFile
$INSTANCE build/3rdparty/ippicv/ippicv_lnx/iw/include/iw/iw_image_op.h cFile
$INSTANCE modules/objdetect/src/aruco/apriltag/apriltag_quad_thresh.hpp cFile
$INSTANCE modules/imgcodecs/src/grfmt_jpegxl.hpp cFile
$INSTANCE modules/gapi/perf/common/gapi_imgproc_perf_tests.hpp cFile
$INSTANCE modules/imgproc/test/test_resize_bitexact.cpp cFile
$INSTANCE modules/core/src/trace.cpp cFile
$INSTANCE 3rdparty/protobuf/src/google/protobuf/descriptor.pb.cc cFile
$INSTANCE modules/imgproc/test/ocl/test_warp.cpp cFile
$INSTANCE modules/dnn/src/torch/THGeneral.cpp cFile
$INSTANCE modules/gapi/include/opencv2/gapi/own/convert.hpp cFile
```

srcml_extraction.ta

File Edit View

```
FACT TUPLE :
$INSTANCE 3rdparty/cpufeatures/cpu-features.h cFile
$INSTANCE 3rdparty/flatbuffers/include/flatbuffers/allocator.h cFile
$INSTANCE 3rdparty/flatbuffers/include/flatbuffers/array.h cFile
$INSTANCE 3rdparty/flatbuffers/include/flatbuffers/base.h cFile
$INSTANCE 3rdparty/flatbuffers/include/flatbuffers/buffer.h cFile
$INSTANCE 3rdparty/flatbuffers/include/flatbuffers/buffer_ref.h cFile
$INSTANCE 3rdparty/flatbuffers/include/flatbuffers/default_allocator.h cFile
$INSTANCE 3rdparty/flatbuffers/include/flatbuffers/detached_buffer.h cFile
$INSTANCE 3rdparty/flatbuffers/include/flatbuffers/flatbuffer_builder.h cFile
$INSTANCE 3rdparty/flatbuffers/include/flatbuffers/flatbuffers.h cFile
$INSTANCE 3rdparty/flatbuffers/include/flatbuffers/stl_emulation.h cFile
$INSTANCE 3rdparty/flatbuffers/include/flatbuffers/string.h cFile
$INSTANCE 3rdparty/flatbuffers/include/flatbuffers/struct.h cFile
$INSTANCE 3rdparty/flatbuffers/include/flatbuffers/table.h cFile
$INSTANCE 3rdparty/flatbuffers/include/flatbuffers/vector.h cFile
$INSTANCE 3rdparty/flatbuffers/include/flatbuffers/vector_downward.h cFile
$INSTANCE 3rdparty/flatbuffers/include/flatbuffers/verifier.h cFile
$INSTANCE 3rdparty/include/opencl/1.2/CL/cl.h cFile
$INSTANCE 3rdparty/include/opencl/1.2/CL/cl.hpp cFile
$INSTANCE 3rdparty/include/opencl/1.2/CL/cl_d3d10.h cFile
```

srcml_extraction.ta include_extraction.ta

File Edit View

```
FACT TUPLE :
$INSTANCE ../../../../local/src/memalloc.h cFile
$INSTANCE ../providers/coreml/coreml_provider_factory.h cFile
$INSTANCE ../providers/dml/dml_provider_factory.h cFile
$INSTANCE 3rdparty/cpufeatures/cpu-features.h cFile
$INSTANCE 3rdparty/flatbuffers/include/flatbuffers/allocator.h cFile
$INSTANCE 3rdparty/flatbuffers/include/flatbuffers/array.h cFile
$INSTANCE 3rdparty/flatbuffers/include/flatbuffers/base.h cFile
$INSTANCE 3rdparty/flatbuffers/include/flatbuffers/buffer.h cFile
$INSTANCE 3rdparty/flatbuffers/include/flatbuffers/buffer_ref.h cFile
$INSTANCE 3rdparty/flatbuffers/include/flatbuffers/default_allocator.h cFile
$INSTANCE 3rdparty/flatbuffers/include/flatbuffers/detached_buffer.h cFile
$INSTANCE 3rdparty/flatbuffers/include/flatbuffers/flatbuffer_builder.h cFile
$INSTANCE 3rdparty/flatbuffers/include/flatbuffers/flatbuffers.h cFile
$INSTANCE 3rdparty/flatbuffers/include/flatbuffers/stl_emulation.h cFile
$INSTANCE 3rdparty/flatbuffers/include/flatbuffers/string.h cFile
$INSTANCE 3rdparty/flatbuffers/include/flatbuffers/struct.h cFile
$INSTANCE 3rdparty/flatbuffers/include/flatbuffers/table.h cFile
$INSTANCE 3rdparty/flatbuffers/include/flatbuffers/vector.h cFile
```

Result of Comparing the TA Files

pairwise_comparison_1.txt

File Edit View

```
=== Understand vs Include ===
Common dependencies: 4777
Exclusive to Understand: 22253
Exclusive to Include: 13439

--- Common dependencies ---
3rdparty/include/opencl/1.2/CL/cl.h cFile
3rdparty/ittnotify/include/ittnotify.h cFile
3rdparty/ittnotify/include/jitprofiling.h cFile
3rdparty/ittnotify/include/legacy/ittnotify.h cFile
3rdparty/ittnotify/src/ittnotify/disable_warnings.h 3rdparty/itt
3rdparty/ittnotify/src/ittnotify/disable_warnings.h cFile
```

pairwise_comparison_3.txt

File Edit View

```
=== Include vs srcML ===
Common dependencies: 3604
Exclusive to Include: 14612
Exclusive to srcML: 0

--- Common dependencies ---
3rdparty/cpufeatures/cpu-features.h cFile
3rdparty/flatbuffers/include/flatbuffers/allocator.h cFile
3rdparty/flatbuffers/include/flatbuffers/array.h cFile
3rdparty/flatbuffers/include/flatbuffers/base.h cFile
3rdparty/flatbuffers/include/flatbuffers/buffer.h cFile
```

pairwise_comparison_2.txt

File Edit View

```
=== Understand vs srcML ===
Common dependencies: 1951
Exclusive to Understand: 25079
Exclusive to srcML: 1653

--- Common dependencies ---
3rdparty/include/opencl/1.2/CL/cl.h cFile
3rdparty/ittnotify/include/ittnotify.h cFile
3rdparty/ittnotify/include/jitprofiling.h cFile
3rdparty/ittnotify/include/legacy/ittnotify.h cFile
3rdparty/ittnotify/src/ittnotify/disable_warnings.h cFile
3rdparty/ittnotify/src/ittnotify/ittnotify_config.h cFile
3rdparty/ittnotify/src/ittnotify/ittnotify_static.h cFile
3rdparty/ittnotify/src/ittnotify/ittnotify_types.h cFile
```

three_way_comparison.txt

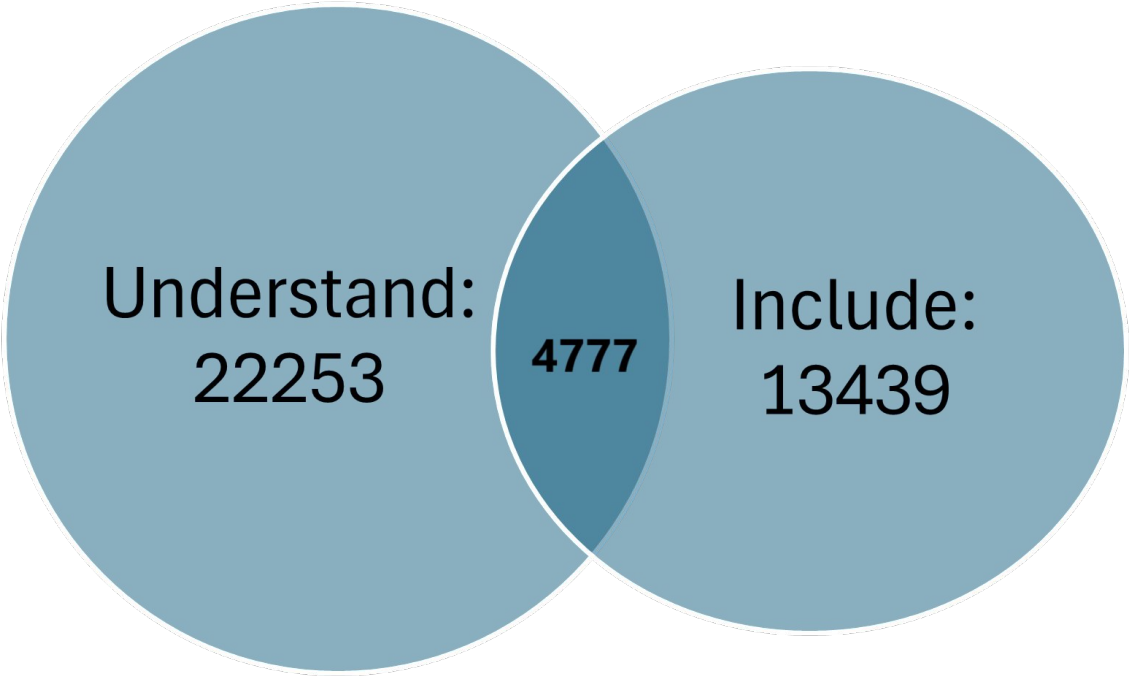
File Edit View

```
=== Three-way comparison: Understand, Include, sr
Common to all three: 1951
Exclusive to Understand: 22253
Exclusive to Include: 11786
Exclusive to srcML: 0

--- Common to all three ---
3rdparty/include/opencl/1.2/CL/cl.h cFile
3rdparty/ittnotify/include/ittnotify.h cFile
3rdparty/ittnotify/include/jitprofiling.h cFile
3rdparty/ittnotify/include/legacy/ittnotify.h cFi
```

```
=== Understand vs Include ===
Common dependencies: 4777
Exclusive to Understand: 22253
Exclusive to Include: 13439

=== Understand vs srcML ===
Common dependencies: 1951
Exclusive to Understand: 25079
Exclusive to srcML: 1653

=== Include vs srcML ===
Common dependencies: 3604
Exclusive to Include: 14612
Exclusive to srcML: 0

=== Three-way comparison: Understand, Include, srcML ===
Common to all three: 1951
Exclusive to Understand: 22253
Exclusive to Include: 11786
Exclusive to srcML: 0
```

Understand Vs. Include

- Total number of dependencies found by these 2 techniques: 40469 dependencies
- Common dependencies: 4777
- Exclusive to Understand: 22253
- Exclusive to Include: 13439
- Understand Coverage:(4777 +22253)/40469 => 66.8%
- Include Coverage: (13439+4777)/40469 => 45%
- Overlap Coverage: 4777/40469=> 11.8%

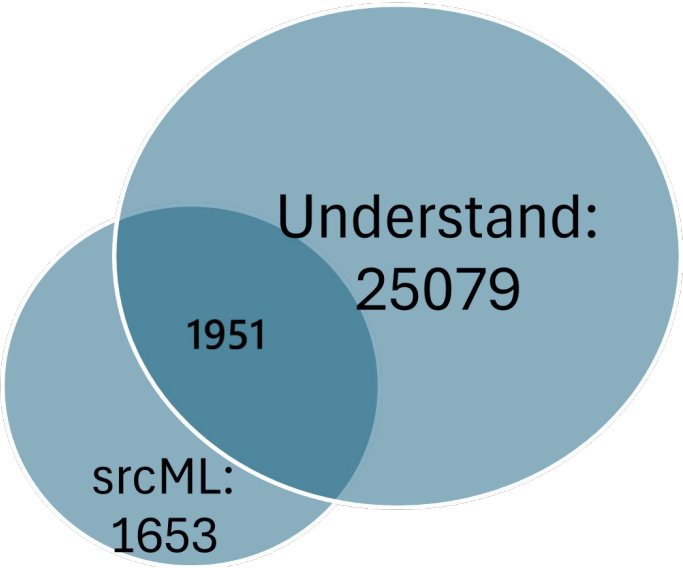


Include Vs. srcML

- Total number of dependencies found by these 2 techniques: 18216 dependencies
- Common dependencies: 3604
- Exclusive to Include: 14612
- Exclusive to srcML: 0
- Include Coverage:100%
- srcML Coverage: 3604/18216 =>19.7%
- Overlap Coverage: 19.7%

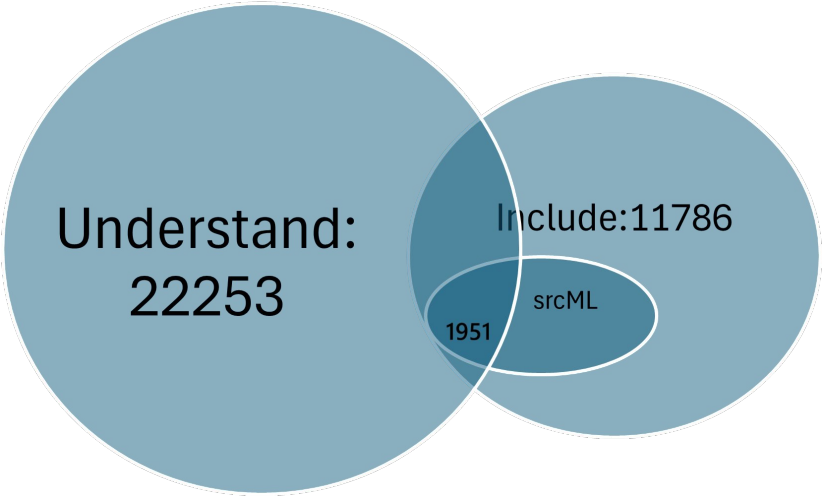
Understand Vs. srcML

- Total number of dependencies found by these 2 techniques:28683 dependencies
- Common dependencies: 1951
- Exclusive to Understand: 25079
- Exclusive to srcML: 1653
- Understand Coverage:27030/28683 => 94.2%
- srcML Coverage: (1653+1951)/28683=>12.6%
- Overlap Coverage: 1951/28683 =>6.8%



Understand Vs. Vs. Include Vs. srcML

- Total number of dependencies found by these 2 techniques: 35990 dependencies
- Common dependencies: 1951
- Exclusive to Understand: 22253
- Exclusive to Include: 11786
- Exclusive to srcML: 0
- Understand Coverage: (22253+1951)/35990 => 67.25%
- Include Coverage: (11786+1951) /35990 => 38.17%
- srcML coverage: (1951)/35990 => 5.42%
- Overlap Coverage: (1951)/35989 => 5.42%



Analysis of Result

Overall Understand reports more dependencies. It performs full semantic analysis and can extract various dependency types:

- function-call relationships
- class inheritance
- type usage
- template instantiation
- variable and constant references
- macro-expanded includes
- fully resolved #include links

Include is syntactic and detects only raw #include directives found in source files including unresolved, redundant, or inactive/commented include statements.

Meanwhile srcML technique converts source code into XML and extracts syntactic elements such as <cpp:include> for include dependencies and <call> for function-call expressions. Unlike Include it doesn't count inactive, macro generated or commented include statements.

Qualitative Analysis

Implications and Metrics(Precision and Recall)

- Precision is used to measure the proportion of correct items [3]
- Recall is used to measure the proportion of retrieved relevant items [3]
- A tradeoff is emphasized between precision and recall [3]
- Higher precision often lowers recall performance [3]
- Metrics elevate dependency extraction technique reliability [3]
- Recall shows completeness [3]
- Precision shows validity [3]

Qualitative Analysis

- Sample size of 380
- Precision = TP/(TP+FP) [3]
- Recall= TP/(TP+FN) [3]
- TP = Common dependencies
- FN = Real dependencies that were missed
- FP = Extra dependencies it incorrectly added

Sample size: **381**

This means 381 or more measurements/surveys are needed to have a confidence level of 95% that the real value is within ±5% of the measured/surveyed value.

Confidence Level: 95%
Margin of Error: 5 %
Population Proportion: 50 % Use 50% if not sure
Population Size: 40469 Leave blank if unlimited population size.
Calculate Clear

Sample size: **380**

This means 380 or more measurements/surveys are needed to have a confidence level of 95% that the real value is within ±5% of the measured/surveyed value.

Confidence Level: 95%
Margin of Error: 5 %
Population Proportion: 50 % Use 50% if not sure
Population Size: 28683 Leave blank if unlimited population size.
Calculate Clear

Sample size: **377**

This means 377 or more measurements/surveys are needed to have a confidence level of 95% that the real value is within ±5% of the measured/surveyed value.

Confidence Level: 95%
Margin of Error: 5 %
Population Proportion: 50 % Use 50% if not sure
Population Size: 18216 Leave blank if unlimited population size.
Calculate Clear

Understand (Reference) Vs. Include (Evaluated)

Total Sample Size = 380

Unique Understand Cases (FN) = 181

Unique Include Cases (FP) = 147

Common Cases (TP) = 52

Precision of Include = 52/(52+147) = 0.261 or 26.1%

Recall of Include = 52/(52+181) = 0.223 or 22.3%

```
=== Processing pairwise_comparison_1.txt ===
  Tools: Understand (reference) vs Include (evaluated)
  Exclusive sections: Understand, Include
  Total edges in file:
    Common: 4777
    Exclusive to Understand: 22253
    Exclusive to Include: 13439
  Sample size: 380
    TP (common): 52
    FN (Understand-only): 181
    FP (Include-only): 147
  Sample written to: understand_vs_include_sample_380.txt
```

Include (Reference) Vs. srcML (Evaluated)

Total Sample Size = 380

Unique Include Cases (FN) = 298

Unique srcML Cases (FP) = 0

Common Cases (TP) = 82

Precision of srcML = 82/(82+0) = 1.00 or 100%

Recall of srcML = 82/(82+298) = 0.216 or 21.6%

```
=== Processing pairwise_comparison_3.txt ===
  Tools: Include (reference) vs srcML (evaluated)
  Exclusive sections: Include, srcML
  Total edges in file:
    Common: 3604
    Exclusive to Include: 14612
    Exclusive to srcML: 0
  Sample size: 380
    TP (common): 82
    FN (Include-only): 298
    FP (srcML-only): 0
  Sample written to: include_vs_srcml_sample_380.txt
```

Understand (Reference) Vs. srcML (Evaluated)

Total Sample Size = 380

Unique Understand Cases (FN) = 333

Unique srcML Cases (FP) = 21

Common Cases (TP) = 26

Precision of srcML = 26/(26+21) = 0.553 or 55.3%

Recall of srcML = 26/(26+333) = 0.072 or 7.2%

```
=== Processing pairwise_comparison_2.txt ===
  Tools: Understand (reference) vs srcML (evaluated)
  Exclusive sections: Understand, srcML
  Total edges in file:
    Common: 1951
    Exclusive to Understand: 25079
    Exclusive to srcML: 1653
  Sample size: 380
    TP (common): 26
    FN (Understand-only): 333
    FP (srcML-only): 21
  Sample written to: understand_vs_srcml_sample_380.txt
```

Analysis of Understand Only Modules

- Understand shows real internal dependencies [6]
  - svm.cpp → parallel.cpp (ML → Core)
  - stitcher.cpp → matchers.hpp (Stitching → Stitching detail)
  - color_hsv.dispatch.cpp → saturate.hpp (ImgProc → Core utilities)
- Resolves semantic relationships that do not appear in direct #include statements
- Provides the most accurate architecture and finds the most meaningful dependencies

understand_vs_include_sample_38

File Edit View

```
# Sampled 380 edges from pairwise_comparison_1.txt
# Tool1 (reference) = Understand
# Tool2 (evaluated) = Include
# Label meanings: common=TP, <Tool1>-only=FN for Tool2, <Tool2>-only=FP for Tool2

Understand-only build/3rdparty/ippicv/ippicv_lnx/iw/src/iw_image_filter_scharr.c build/3rdparty/ippicv/ippicv_lnx/icv/include/ippicv_types_l.h
Understand-only modules/ml/src/svm.cpp modules/core/src/parallel.cpp
Include-only    3rdparty/protobuf/src/google/protobuf/stubs/mutex.h 3rdparty/protobuf/src/google/protobuf/port_def.inc
Understand-only modules/calib3d/src/p3p.cpp modules/calib3d/src/polynom_solver.cpp
Understand-only modules/calib3d/src/calibinit.cpp modules/imgproc/src/convhull.cpp
common  modules/dnn/src/net_impl.hpp modules/dnn/src/op_webnn.hpp
common  modules/features2d/test/test_utils.cpp cFile
Understand-only modules/stitching/src/motion_estimators.cpp modules/calib3d/src/calibration_base.cpp
```

Analysis of Include Only Modules

- Include only samples largely consist of noise
  - sample/demo code (samples/cpp/...)
  - 3rd-party libraries (protobuf, libjasper, openexr)
  - system/STL headers (cmath, iostream)
- Records every raw #include even those irrelevant to architecture
- Produces a lot of noise and the lowest-quality architectural graph

understand_vs_include_sample_38

File Edit View

```
# Sampled 380 edges from pairwise_comparison_1.txt
# Tool1 (reference) = Understand
# Tool2 (evaluated) = Include
# Label meanings: common=TP, <Tool1>-only=FN for Tool2, <Tool2>-only=FP for Tool2

Understand-only build/3rdparty/ippicv/ippicv_lnx/iw/src/iw_image_filter_scharr.c build/3rdparty/ippicv/ippicv_lnx/icv/include/ippicv_types_l.h
Understand-only modules/ml/src/svm.cpp modules/core/src/parallel.cpp
Include-only     3rdparty/protobuf/src/google/protobuf/stubs/mutex.h 3rdparty/protobuf/src/google/protobuf/port_def.inc
Understand-only modules/calib3d/src/p3p.cpp modules/calib3d/src/polynom_solver.cpp
Understand-only modules/calib3d/src/calibinit.cpp modules/imgproc/src/convhull.cpp
common  modules/dnn/src/net_impl.hpp modules/dnn/src/op_webnn.hpp
common  modules/features2d/test/test_utils.cpp cFile
Understand-only modules/stitching/src/motion_estimators.cpp modules/calib3d/src/calibration_base.cpp
Include-only    cmath cFile
```

Analysis of srcML Only Modules

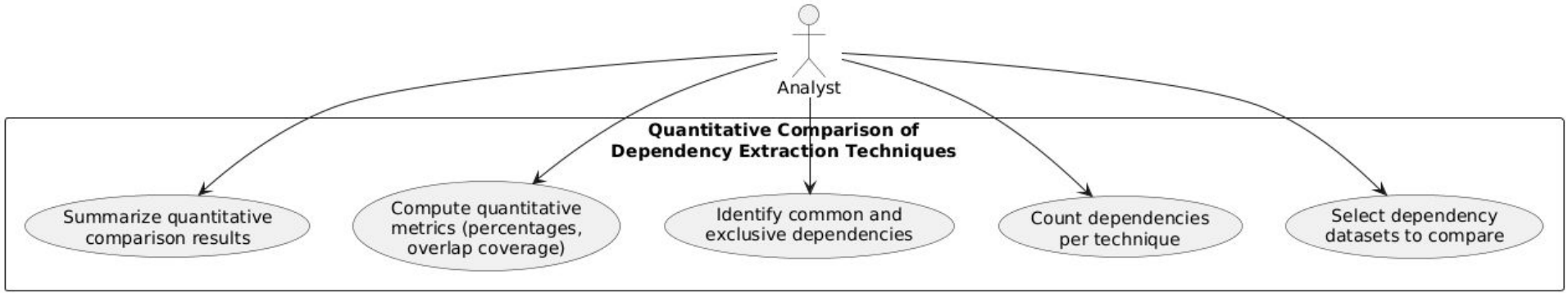
- srcML mostly references:
  - 3rd-party libraries
  - backend optimization layers (carotene HAL)
- Analyzes Abstract Syntax Tree (AST) structure not semantic relationships missing cross-module dependencies [5]
- Very shallow captures far fewer dependencies and misses most meaningful ones

understand_vs_srcml_sample_380

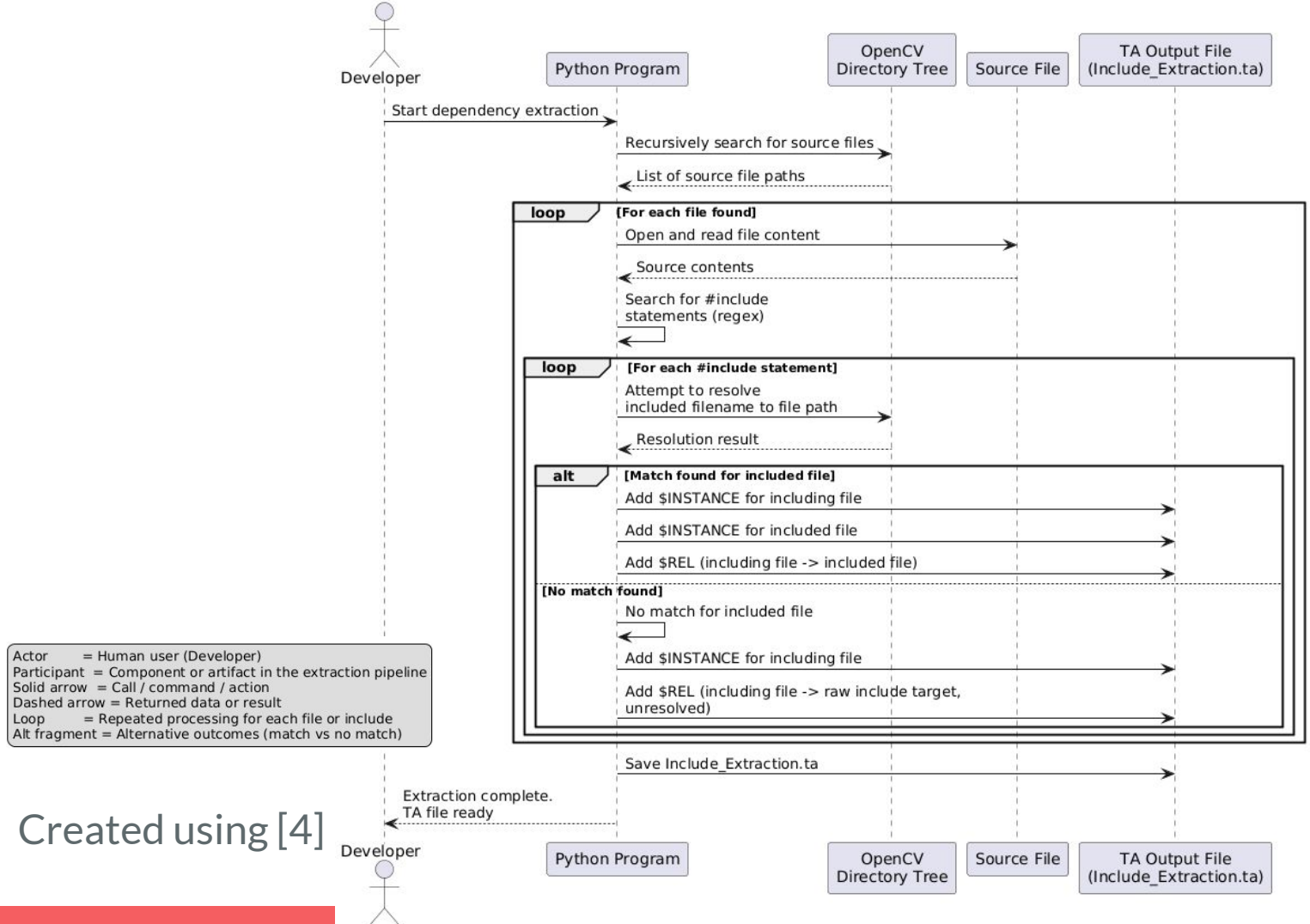
File Edit View

```
# Sampled 380 edges from pairwise_comparison_2.txt
# Tool1 (reference) = Understand
# Tool2 (evaluated) = srcML
# Label meanings: common=TP, <Tool1>-only=FN for Tool2, <Tool2>-only=FP for Tool2

Understand-only modules/imgcodecs/test/test_animation.cpp modules/core/include/opencv2/core/mat.inl.hpp
srcML-only      hal/carotene/src/add_weighted.cpp cFile
Understand-only modules/stitching/src/seam_finders.cpp modules/imgproc/include/opencv2/imgproc/detail/gcgraph.hpp
Understand-only modules/imgcodecs/src/grfmt_exr.cpp modules/core/src/convert.dispatch.cpp
Understand-only modules/dnn/misc/tensorflow/tensor_shape.pb.h 3rdparty/protobuf/src/google/protobuf/message.cc
Understand-only modules/dnn/src/layers/detection_output_layer.cpp modules/core/src/convert.dispatch.cpp
Understand-only modules/photo/test/test_npr.cpp modules/ts/src/ts_func.cpp
srcML-only      samples/cpp/image_alignment.cpp cFile
Understand-only modules/dnn/src/net.cpp modules/core/src/system.cpp
```

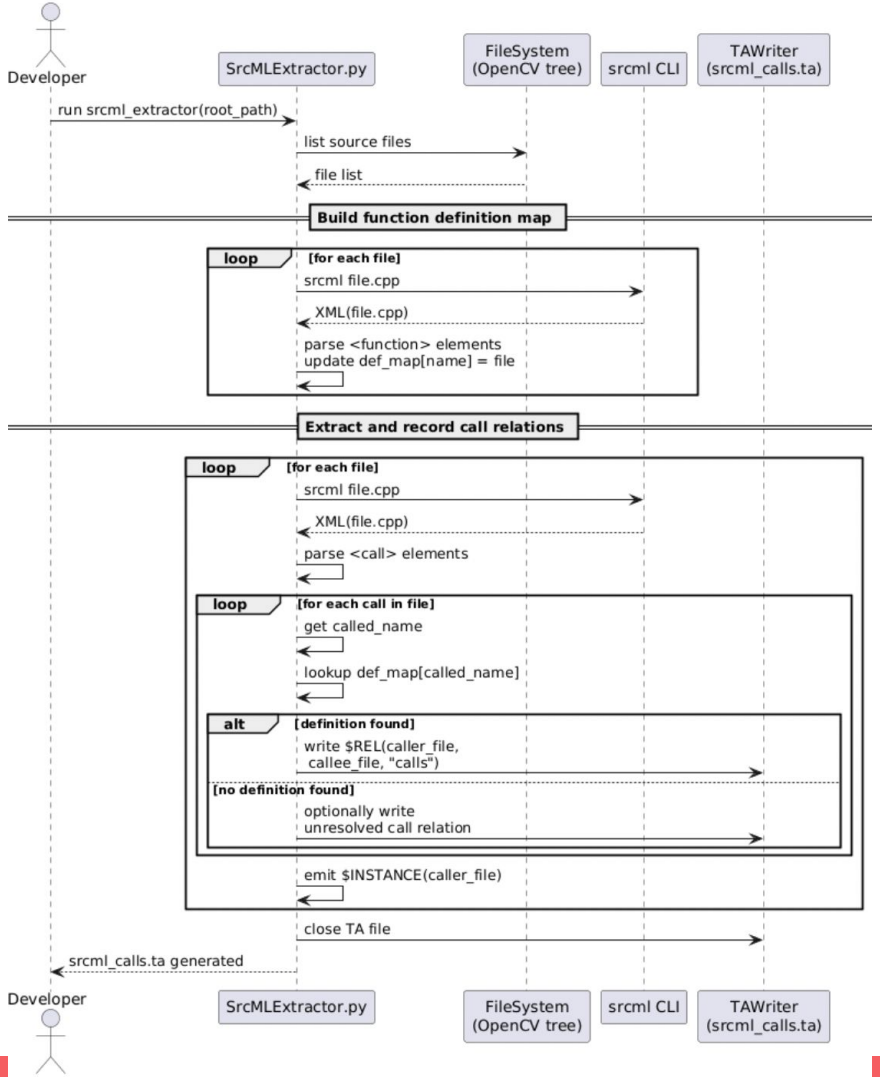
Use Case



Created using [4]



Created using [4]



Actor = Human user (Developer)
Participant = Component or artifact in the extraction pipeline
Solid arrow = Call / command / action
Dashed arrow = Returned data or result
Loop = Repeated processing for each file or include
Alt fragment = Alternative outcomes (match vs no match)

Created using [4]

Risks & Limitations Of Reported Findings

Understand

- Treated as reference, but not perfect.
- Can misinterpret dependencies involving templates, macros, and hardware-specific dispatch
- Output varies with build configuration (flags, includes).
- Largest graph ≠ true architecture in all environments.

Include-Based Extraction

- Purely syntactic: records literal #include only.
- High noise from samples, tests, vendor code, and system headers.
- Filename matching can incorrectly link files across modules.
- Ignores macro logic and conditional compilation → misses real dependencies.

srcML Extraction

- Structural but not semantic → cannot resolve overloads, templates, or cross-module usage.
- Misses many meaningful dependencies → smaller graph.
- Often captures 3rd-party/backend code due to simpler syntax.
- Slow on large projects → risk of partial extraction.

General Limitations

- Results depend on tool behaviour, OpenCV's large/complex codebase, and comparison method.
- Completeness and accuracy vary across extraction techniques.

Comparison Methodology

- TA format normalization loses dependency types.
- Precision/recall based on Understand → biased metrics.
- Sampling 380 edges introduces uncertainty.
- Vendor libraries add noise and distort overlap comparisons.

Overall Risk

- No method fully captures OpenCV's architecture.
- Understand over-resolves.
- Include over-approximates
- srcML under-approximates.

Lessons Learned

- The dependency extraction tools greatly vary in dependency coverage
- Understand gives the strongest semantically dependency results
- Include is very simple however it is limited structurally
- srcML catches structure and function calls
- What recall and precision do is assess extraction reliability effectively
- The differences between the tools make interpretation and accuracy limits

References

[1] “Understand: The Software Developer’s Multi-Tool,” *Scitools.com*, 2024. https://scitools.com/

[2] “srcML,” *Srcml.org*, 2025. https://www.srcml.org/ (accessed Nov. 19, 2025).

[3] “Precision and Recall,” Accessed: Nov. 22, 2025. [Online]. Available: https://www.britannica.com/science/precision-and-recall

[4] “PlantUML,” PlantUML.com, https://plantuml.com/ (accessed Oct. 28, 2025).

[5] “srcML – Wikipedia.” Accessed: Nov. 25, 2025. [Online]. Available: https://en.wikipedia.org/wiki/SrcML

[6] “Understand – SciTools User Guide and Reference Manual.” Accessed: Oct. 10, 2025. [Online]. Available: https://docs.scitools.com/manuals/pdf/understand.pdf

Thank you for Listening!